

Smart Library

An IoT System for Environmental Monitoring and Occupancy
Analysis

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The Need for a Smart Library

- Research and real world evidence shows that environmental conditions strongly influence students' learning experiences.



Lighting levels

Affect students' mood and ability to stay focused.



Noise levels

Are one of the main factors students consider when choosing where to study.



Air quality

Influence comfort and concentration, and also impacts productivity and overall comfort.



Space availability

Plays a role in how effectively students can work.

Project Goal & Justification

Goal: To develop an IoT system that monitors environment and occupancy.

This system provides actionable insights for:



Improving User Comfort

Ensure a pleasant and productive environment for all patrons.



Optimizing Space

Make data-driven decisions about layout and resource allocation.



Reducing Energy Consumption

Align HVAC and lighting with actual library usage to cut costs.

UN Sustainable Development Goals

Goal 11: Sustainable Cities

By making a public building like a library more efficient, safe, and comfortable, the project contributes to the sustainability of urban community infrastructure. It helps create smarter public spaces that respond to user needs.

Goal 7: Affordable & Clean Energy

The system provides the data necessary to reduce energy waste. By correlating occupancy with energy usage, the library can implement strategies (like automated lighting/HVAC) to lower consumption and improve energy efficiency.

Users and Their Stories



Library Administrator

"As an Administrator, I want to see hourly and daily occupancy trends so that I can optimize staffing schedules and allocate resources."



Facility Manager

"As a Facility Manager, I want to monitor temperature and air quality so that I can ensure a comfortable environment and perform proactive HVAC maintenance."



Students

"As a student, I want to check how crowded the library is before I visit so that I can choose the best time to go and find a study space."

Project Management Approach

A modular and iterative approach was used for solo development:



Task Breakdown

Project was broken into key components: Data Collection, Communication, Storage, and Visualization.



Centralized Documentation

A central file was maintained to document architecture, setup instructions, and development progress.



Simulation-First Development

A simulation file was created to allow parallel development and testing of the backend without the physical hardware.



The Technology Stack

An overview of the hardware, software, and data
flow.

Technologies Used: Hardware



Raspberry Pi

The central controller and brain of the operation, running the data collection script.



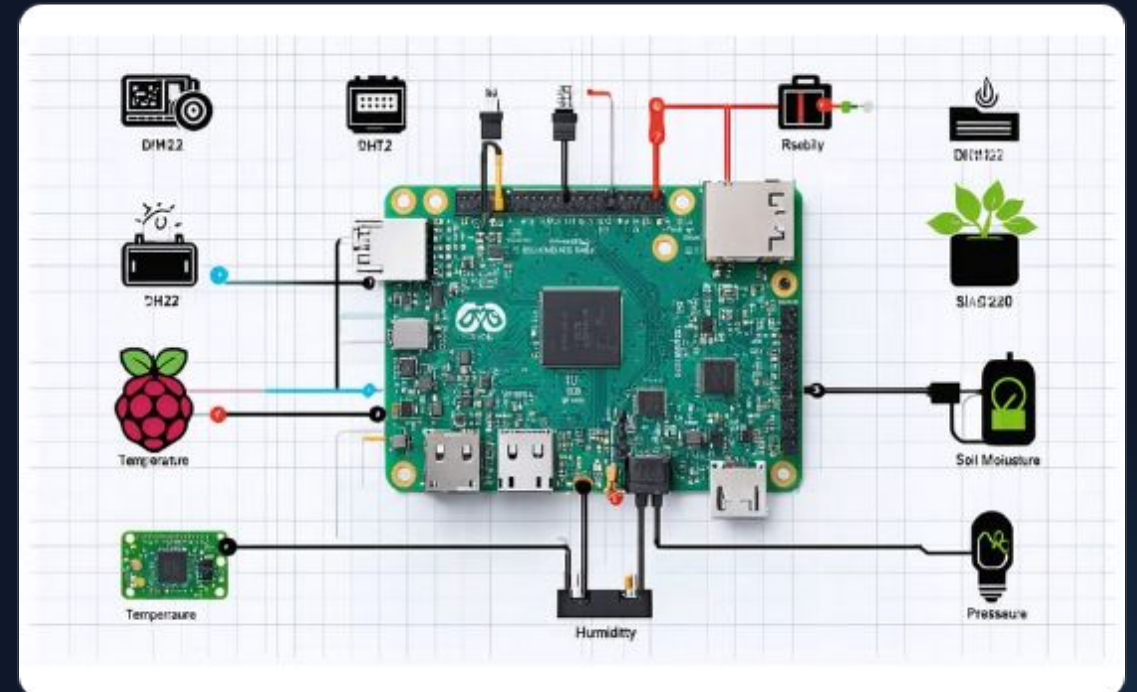
Arducam Camera

Provides the video feed for person detection and calculating ambient light levels.



BME680 Sensor

An all-in-one sensor that collects Temperature, Humidity, Pressure, and Air Quality (IAQ) data.



Technologies Used: Software



Python: Main language for data collection and simulation.



Docker & Docker Compose: For containerizing and orchestrating all backend services.



InfluxDB: Time-series database for efficiently storing all sensor data.



Grafana: For all data visualization and dashboarding.



Mosquitto (MQTT): Lightweight messaging protocol for IoT communication.



Telegraf: Agent for collecting, processing, and writing data to InfluxDB.

Technologies Used: Libraries & APIs

Key Libraries

- **OpenCV (Python):** For image processing (light level) and person detection.
- **Paho-MQTT (Python):** Client library for connecting to the MQTT broker.
- **bme680 (Python):** Library for interfacing with the BME680 sensor.

Models & Queries

- **Flux:** Query language used in Grafana to retrieve and process data from InfluxDB.
- **MobileNet-SSD:** A pre-trained deep learning model optimized for running on low-power devices like a Raspberry Pi.

Software & Hardware Architecture



1. Collect

pi_controller.py on the Raspberry Pi collects sensor data and detects people using the camera.



2. Communicate

The script publishes this data as a JSON payload to the Mosquitto MQTT broker.



3. Store

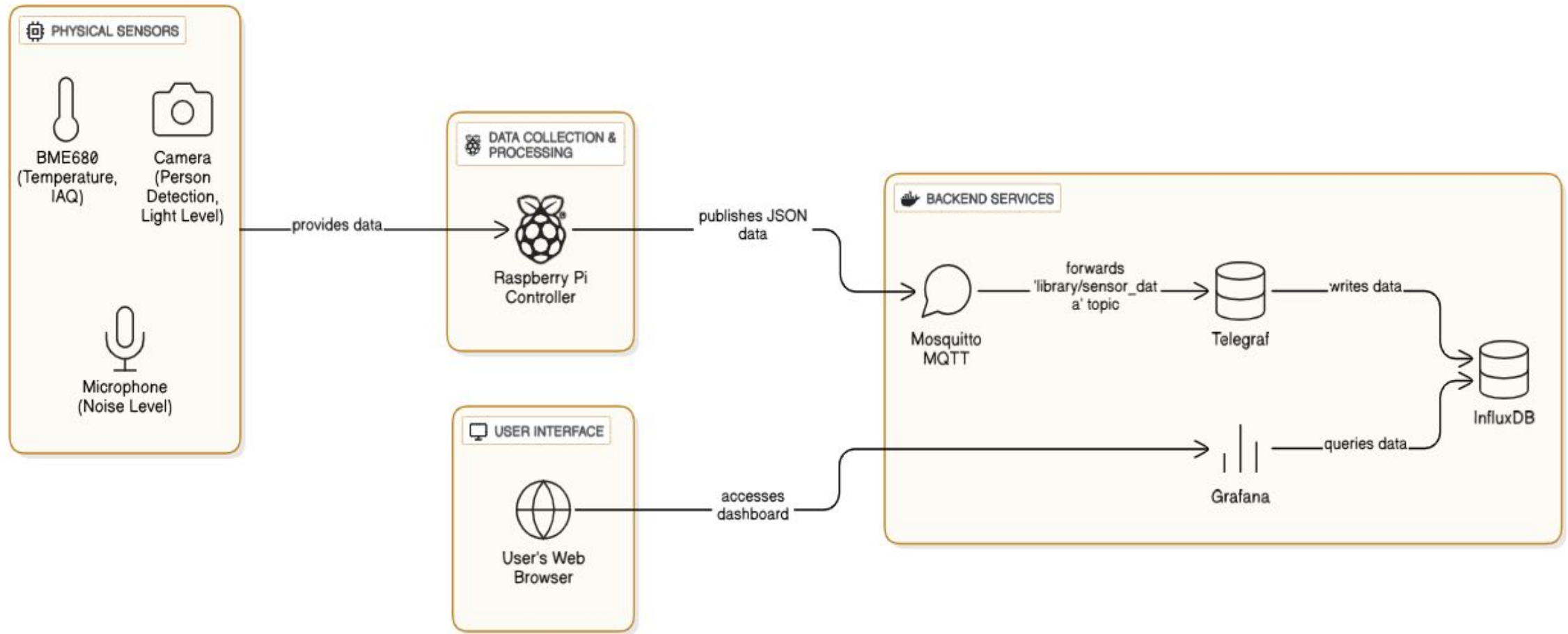
Telegraf subscribes to the MQTT topic, parses the JSON, and writes it into the InfluxDB database.



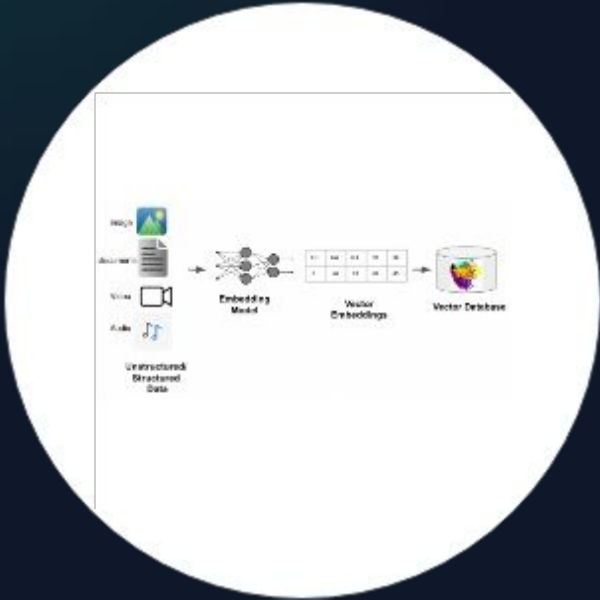
4. Visualize

Grafana connects to InfluxDB as a data source and displays the data on a real-time dashboard.

Software & Hardware Architecture



Database Architecture (InfluxDB)



Measurement: Sensor Readings

Tags (for indexing):

location (e.g., "library_zone_A")

sensor_id (e.g., "pi_controller_01")

Fields (the actual values):

people_count (integer)

temperature (float)

iaq (float),

light_level (float),

noise_db (float)

Environmental Impact

Positive Impact

Energy Reduction: Provides the data needed to intelligently control lighting and HVAC systems based on actual occupancy and environmental conditions.

Resource Optimization: Better management of library spaces and staff leads to more efficient use of existing resources.

Negative Impact

E-Waste: The project introduces electronic hardware (Raspberry Pi, sensors) which has a limited lifespan and will eventually become electronic waste.

Energy Consumption: The Raspberry Pi and the server running the backend services consume electricity. While low-power, the cumulative effect is non-zero.

Ecodesign Evaluation (RGESN)

Global Progress Score	63 %
Date of evaluation	15/11/2025
Score by theme :	
1 – Strategy (details page 2 and page 3)	93 %
2 – Specifications (details page 4 and page 5)	100 %
3 – Architecture (details page 6)	36 %
4 – User Experience and Interface (details page 7 and page 8)	31 %
5 – Content (details page 9)	0 %
6 – Frontend (details page 10)	47 %
7 – Backend (details page 11)	36 %
8 – Hosting (details page 12 and page 13)	79 %
9 – Algorithmics (details page 14)	79 %

Main Compliant Criteria (Our Strengths)



High Utility & Strategy

The dashboard meets a validated user need (finding study space) without superfluous features, avoiding digital waste.



Data Responsibility

We collect only essential signals (occupancy counts), avoiding personal tracking, facial recognition, or metadata.



Frugal User Interface

The design avoids "heavy" media—no autoplay videos, no infinite scrolling, and no complex animations.



Hardware Compatibility

The service works on older devices and low-speed connections thanks to its lightweight architecture and standard web technologies.



Open Source Stack

Built entirely on open-source technologies (InfluxDB, Grafana, Mosquitto), ensuring transparency, longevity, and no vendor lock-in.



Efficient Storage Strategy

We use a Time-Series Database (TSDB) which compresses sensor data far more efficiently than traditional relational databases.

Main Non-Compliant Criteria (Roadmap)



Data Lifecycle

We must configure strict Retention Policies in InfluxDB to automatically delete raw sensor data older than 30 days to prevent disk bloat.



Protocol Optimization

Currently sending JSON text over MQTT. Switching to a binary format (like Protobuf) would reduce payload size for network communication.



Impact Documentation

We need to formally calculate and document the energy ROI:
Does the energy saved by students finding seats outweigh the server's run cost?



UX Enhancements

Implement a "Viability Score" or similar single metric to simplify decision-making for users and reduce analysis paralysis when selecting a study area.



Energy Monitoring

We lack hardware tools (like smart plugs) to measure the actual real-time power consumption of the Raspberry Pi.

Opportunities for Improvement

Short / Mid-Term

- **Implement Smart Controls:** Integrate with smart plugs or building APIs to automatically dim lights or adjust the thermostat.
- **Optimize Data Collection:** Make the collection frequency dynamic (e.g., publish data less frequently).
- **Event-Driven Detection:** Run the person detection model only when motion is first detected.

Long-Term

- **Sustainable Hardware Sourcing:** When scaling, choose hardware from manufacturers with strong environmental commitments.
- **Dynamic Forecasting:** Develop advanced predictive models to forecast space utilization beforehand.
- **Holistic Lifecycle Management:** Develop a plan for device repairs, upgrades, and responsible recycling.

REFERENCES

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Questions?

Thank you for your attention.